MBA Fastrack 2025 (CAT + OMETs)

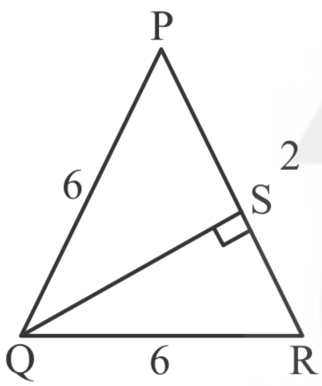
QUANTITATIVE APTITUDE

2D Geometry - 1

DPP: 1

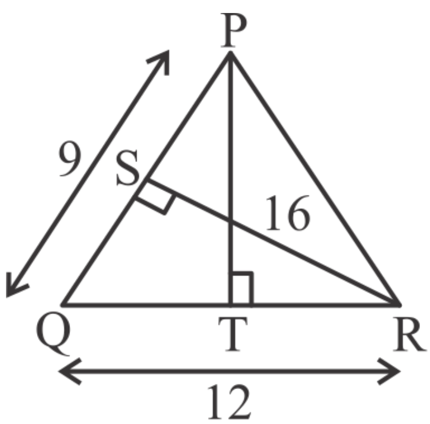
Q1 The base of a triangle is 9 cm and the height is 15 cm. The height (in cm) of another triangle, having base 15 cm and 4 times the area of the first triangle, is:

Q2 Find the length of altitude (in cm) to side PR of a triangle PQR with side PQ = QR = 6 cm and PR = 2 cm.



(A) $\sqrt{15}$ (B) $\sqrt{21}$
(C) $\sqrt{33}$ (D) $\sqrt{35}$

Q3 In the following figure, PQ = 9 cm, QR = 12 cm and RS = 16 cm. Then, the length of PT is how much percentage less than the length of RS given that $PT \perp QR$ and $RS \perp PQ$.



Q4 An isosceles triangle has two sides of lengths 5 cm and 6 cm, and its area is 12 cm^2. What is the length of the third side (in cm)?
(A) 5 (B) 6
(C) 5 or 6 (D) None of these

Q5 If the sides of an equilateral triangle become 3 times the original sides, then by what percentage does its area increase?

Q6 The perimeter of an isosceles triangle is 60 cm, the length of one of the sides is 20 cm, and the angle between the equal sides is x^o. Find the value of x.

Q7 Two sides, other than the longest side, of a triangle forming a right angle are x^2 units and $4x^2 - 16$ units. If the area of the triangle is 24 square units, then what is the perimeter of the triangle (in units)?

Q8 The ratio of length of sides of a triangle is 8 : 15 : 17. If the area of the triangle is 240 square units, then the length (in units) of the largest side is:

Q9 Find the maximum possible area $\left(\text{in } \ \mathrm{cm}^2\right)$ of an isosceles triangle PQR having Perimeter as 18 cm and length of one of the sides as 8 cm.
(A) $3\sqrt{7}$ (B) 12
(C) $7\sqrt{3}$ (D) 8

Q10 In triangle ABC, the measure of B is 45°, and the lengths of sides AB and AC are $4\sqrt{2}$ cm and 5 cm, respectively. Find the area of triangle ABC in sq. cm.

Q11 Let in a triangle PQR, S is any point on QR such that QS : SR = 5 : 1. T is any point on the line PS such that PT : TS = 3 : 2. Find the area of the triangle PQT (in cm^2) if the area of the triangle PQR is 36 cm^2.

Q12 A triangle has positive integer side lengths a, b, and c such that a - b = 4 units, a + b + c = 24 units and a > b > c. If the triangle is obtuse, find its area (in sq. units).

(A) $4\sqrt{10}$ (B) $5\sqrt{10}$
(C) $6\sqrt{10}$ (D) $7\sqrt{10}$

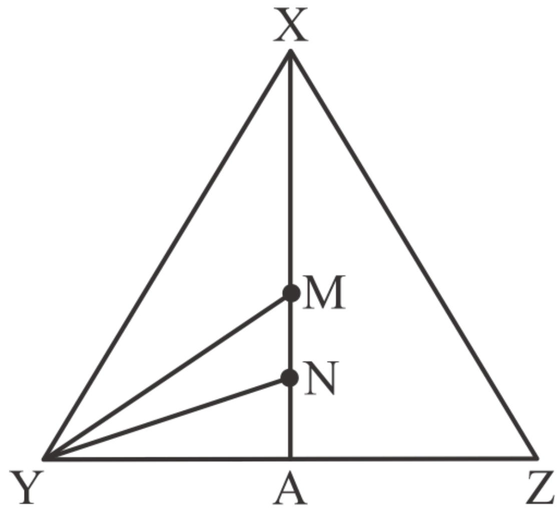
Q13 Let PQR be a right-angled triangle right-angled at P with hypotenuse QR of length 40 cm. If PS is perpendicular to QR, then the maximum possible length of PS, in cm, is

Q14 M is the incenter of PQR. Given that PMR=104° and QMR=110°, determine the measure(in degrees) of PRQ.

Q15 If the area of the circumcircle of an equilateral triangle is 100 sq. units, then find the area (in sq. units) of the incircle of the same equilateral triangle

Q16 PQRS is a rectangle with sides PQ = 42 and QR = 72, and T is the midpoint of the side RS. Then, the length, of radius of incircle of $\triangle$PST is

Q17 In the given figure, XA is the median of triangle XYZ, with M as the midpoint of XA and N as the centroid of the triangle. What is the ratio of the area of triangle XYZ to the area of triangle YMN?



(A) 6:1 (B) 9:1
(C) 12:1 (D) 18:1

Q18 PQRS is a square with each side measuring 12 cm. PX bisects QPR, where X is a point on QR. Determine the area of PRX.

(A) $72\left(\sqrt{2}-1\right)\ cm^2$
(B) $72\left(2-\sqrt{2}\right)\ cm^2$
(C) $\frac{72}{\sqrt{2}-1}\ cm^2$
(D) $\frac{72}{2-\sqrt{2}}\ cm^2$

Q19 Three villages, X, Y, and Z, are located in the form of a triangle such that the distance between X and Y is 10 km, between X and Z is also 10 km, and between Y and Z is 16 km. A communication tower is built at a point that is equidistant from all three villages. Find the distance from the tower to any of the villages.

(A) 8 km (B) $8\frac{1}{3}\ km$
(C) $\frac{8}{3}\ km$ (D) $4\sqrt{2}\ km$

Q20 In PQR, medians PD, QE, and RF are drawn from vertices P, Q, and R respectively. Given that PQ=10 cm, QR=12 cm, and PR=16 cm, determine the value of $\mathrm{PD}^2+\mathrm{QE}^2+\mathrm{RF}^2$ in square centimeters.

Q21 In XYZ, M is the midpoint of the median from X. If the area of MYZ is 15 square units, find the area of NYZ in square units, where N is the centroid of the triangle.

Q22 In PQR, PM is a median with M on QR, and QN is an angle bisector of PQR with N on PR. PM and QN intersect at O. If the ratio PO:OM is 4:3, find the ratio PN:NR.

(A) 1:2 (B) 3:4

(C) 1:1 (D) 2:3

Q23 In PQR, it is given that POQ=110^0 and QSR=152^0. Determine the measure of PIR in degrees, where O and S represent the orthocentre and circumcentre of the triangle, respectively, and I is its incentre

Q24 In PQR, the medians PX and QY intersect at a right angle. Given that QY measures 18 units and the total area of PQR is 156 square units, find the length of PX in units.

Q25 In a right-angled triangle PQR, where Q is the right angle and PQ is smaller than QR, the angle bisector and median drawn from Q intersect PR at S and T, respectively. If the distance between S and T is 3.5 cm and PR is 35 cm, find the area of triangle PQR rounded off to the nearest integer.(in square cm).

Q26 In an isosceles triangle XYZ with XY = XZ, the perpendicular drawn to side YZ from the opposite vertex is 15 units long. Additionally, the median drawn to side XZ from the opposite vertex has a length of 19.5 units. Determine the area of triangle XYZ.

(A) 150 sq units (B) 180 sq units

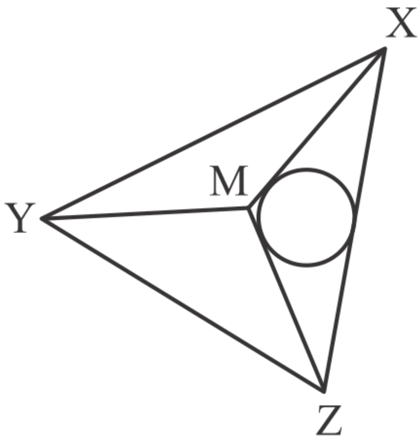
(C) 195 sq units (D) 292.5 sq units

Q27 In triangle PQR, PQ = 16 units and QR = 12 units. The medians from vertices R and P intersect at a right angle inside the triangle. Determine the length of PR.

(A) $4\sqrt{2}$ units (B) $4\sqrt{3}$ units

(C) $4\sqrt{5}$ units (D) 8 units

Q28 XYZ is an equilateral triangle with a side length of $4\sqrt{3}$ cm. M is the centroid of the triangle. A circle is inscribed within MXZ. Determine the radius of this circle.



(A) $(4\sqrt{3} - 6)\ cm$

(B) $(4 - 2\sqrt{3})\ cm$

(C) $(2\sqrt{3} - 1)\ cm$

(D) $\frac{2\sqrt{3}}{3}\ cm$

Q29 A point X is lying inside an equilateral triangle such that the sum of perpendicular distances of X from all three sides is $8\sqrt[4]{3}$ cm. What will be the area of the triangle in square centimeters?

Q30 If A is any point lying inside a right angled isosceles triangle PQR right angled at Q. If the shortest distance of point A from all the three sides of the triangle is $(6 - 3\sqrt{2})$ cm, then what is the area of the triangle in square centimeters?

Answer Key

Q1	36
Q2	D
Q3	25
Q4	A
Q5	800
Q6	60
Q7	24
Q8	34
Q9	B
Q10	14
Q11	18
Q12	C
Q13	20
Q14	112
Q15	25
Q16	9
Q17	C
Q18	B
Q19	B
Q20	375
Q21	10
Q22	D
Q23	107
Q24	13
Q25	283
Q26	B
Q27	C
Q28	A
Q29	64
Q30	18

Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

Given:

Base of a triangle = 9 cm and height = 15 cm

Therefore, area of triangle = $\frac{1}{2}$ × 9 × 15 = $\frac{135}{2}$ cm^2

Now, the area of another triangle = 4 × $\frac{135}{2}$ = 270 cm^2

The height of the other triangle = $\frac{270}{\frac{1}{2}\times15} = 36$ cm

Video Solution:

Q2 Text Solution:

Since we know all the lengths of the sides, therefore, the semi-perimeter of the triangle PQR = $\frac{6+2+6}{2} = 7$ cm

Thus, by the Heron's formula, we have the area of the triangle PQR

$$= \sqrt{7(7-6)(7-6)(7-2)}$$

$$= \sqrt{7\times1\times1\times5}$$

$$= \sqrt{35}\, cm^2$$

Now, again, since, QS is the altitude to the side PR, therefore, the area of the triangle PQR

$$= \frac{1}{2}\times PR\times QS = \frac{1}{2}\times QS\times 2 = \sqrt{35}$$

So, QS = $\sqrt{35}$ cm

Video Solution:

Q3 Text Solution:

Since, $RS \perp PQ$ and PQ = 9 cm, RS = 16 cm, so the area of the triangle PQR

= $\frac{1}{2}\times 16\times 9\ \mathrm{cm}^2$

Again, since, $PT \perp QR$ and QR = 12 cm, so the area of the triangle PQR

= $\frac{1}{2}\times PT\times 12\ \mathrm{cm}^2$

Thus, equating the values of the two areas of the triangle PQR, we get

$$\frac{1}{2}\times16\times9 = \frac{1}{2}\times PT\times12$$

$$PT = \frac{16\times9}{12} = 12\ \mathrm{cm}$$

So, the required % =

$$\frac{RS-PT}{RS}\times100 = \frac{16-12}{16}\times100 = 25\%$$

Video Solution:

Q4 Text Solution:

Case 1 (Third side = 5 cm):

Sides: 5, 5, 6.

Semi-perimeter $s = \frac{5+5+6}{2} = 8$ cm

Area of the triangle =

$$\sqrt{8\cdot(8-5)\cdot(8-5)\cdot(8-6)}$$

$$= \sqrt{8\cdot3\cdot3\cdot2} = 12$$

Case 2 (Third side = 6 cm):

Sides: 5, 6, 6.

Semi-perimeter $s = \frac{5+6+6}{2} = \frac{17}{2}$

Area of the triangle

$= \sqrt{\frac{17}{2}\left(\frac{17}{2}-5\right)\left(\frac{17}{2}-5\right)\left(\frac{17}{2}-6\right)} \neq 12$ (invalid).

Thus, the only possible value of the third side is 5 cm.

Video Solution:

Q5 Text Solution:

We know,

a rea of an equilateral triangle $=$, $\frac{\sqrt{3}}{4}a^2$

where a is the side of the equilateral triangle

Let $A_1 = \frac{\sqrt{3}}{4}a^2$, then,

$A_2 = \frac{\sqrt{3}}{4} \times (3a)^2 = \frac{\sqrt{3}}{4} \times 9a^2$

Now,

$$\frac{A_2}{A_1} = \frac{\frac{\sqrt{3}}{4} \times 9a^2}{\frac{\sqrt{3}}{4} \times a^2} = \frac{9}{1} = \frac{9x}{x}$$

Therefore, the area of the equilateral triangle is increased by

=

$$\left(\frac{A_2 - A_1}{A_1}\right) \times 100\% = \left(\frac{9x - x}{x}\right) \times 100\% = \frac{8x}{x} \times 100\% = 8 \times 100\% = 800\%$$

Video Solution:

Q6 Text Solution:

Given that, the perimeter of the isosceles triangle is 60 cm, where one of its side lengths is 20 cm.

Therefore, we have only the possibility to get the lengths of the sides as, 20 cm, 20 cm and 20 cm.

So, the triangle turns out to be equilateral. (We know that equilateral triangles are also isosceles triangles)

Thus, all the angles of it are 60°.

Hence, the only possible value of x is 60.

Video Solution:

Q7 Text Solution:

Given that, two sides of a triangle forming a right angle are x^2 and $4x^2 - 16$.

Area of triangle will be:

$\frac{1}{2} \times x^2 \times (4x^2 - 16) = 24$ (Given)

Let $x^2 = y$. Then, we have

$$\frac{1}{2} \times y \times (4y - 16) = 24$$

$$y(y-4) = 12$$

$$y^2 - 4y - 12 = 0$$

$$(y-6)(y+2) = 0$$

$$y = 6 \ [\text{since, } -\text{ve value can't be taken here}].$$

i.e., $x^2 = 6$

Then, the values of the smaller sides of the right-angled triangle are: 6, $4 \times 6 - 16$, i.e., 6, 8

The values (6, 8, 10) are the triplets of a right-angled triangle.

Then, the length of the hypotenuse is 10 units.

Perimeter will be: 6 + 8 + 10 = 24 units.

Video Solution:

Q8 Text Solution:

Let the sides of the triangles be: 8x, 15x and 17x.

Therefore, the semi-perimeter of the triangle

= $\frac{8x+15x+17x}{2} = 20x$

Hence, the area of the triangle

=

$$\sqrt{20x(20x-8x)(20x-15x)(20x-17x)}$$

$$= \sqrt{20x \times 12x \times 5x \times 3x} = 60x^2$$

= 240 (Given)

$\Rightarrow x^2 = \frac{240}{60} = 4$

$\Rightarrow x = 2$

Thus, the length of the largest side = 17(2) = 34 units.

Alternatively,

We can clearly notice that 8, 15 and 17 forms a right triangle.

Let the sides be 8x, 15x and 17x

Area = $\frac{1}{2} \times 8\mathrm{x} \times 15\mathrm{x} = 60\mathrm{x}^2$

$60\mathrm{x}^2 = 240$

$\mathrm{x}^2 = 4$

$\mathrm{x} = 2 \left(\mathrm{x \ cannot \ be} \ -2, \ \mathrm{as \ the \ length \ cannot \ be \ negative}\right)$

Thus, the longest side = 17x = 34 units

Video Solution:

Q9 Text Solution:

Since, the perimeter of the isosceles triangle is given as 18 cm and one side length is given as 8 cm, so there are two possibilities for obtaining the side lengths:

Case 1: The side lengths are: 8, 8, 2

The semi-perimeter of the triangle =

$\frac{8+8+2}{2} = 9$ cm

Thus, the area of the triangle =

$$\sqrt{9(9-8)(9-8)(9-2)}$$

$$= \sqrt{9 \times 1 \times 1 \times 7} = 3\sqrt{7} \ \mathrm{cm}^2$$

Case 2: The side lengths are: 8, 5, 5

We need not require to calculate the semi-perimeter again since, the perimeter is the same in each case.

So, in this case, the area of the triangle =

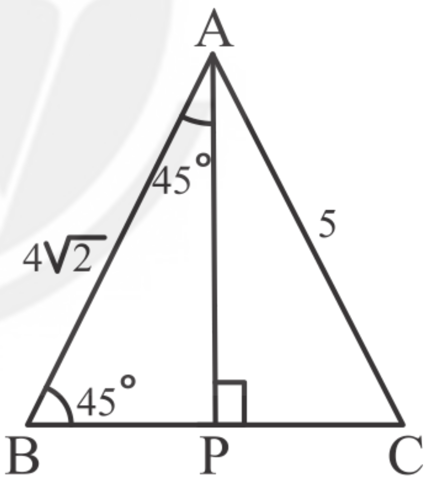
$\sqrt{9(9-8)(9-5)(9-5)}$ $> 3\sqrt{7} \ \mathrm{cm}^2$.

$$= \sqrt{9 \times 1 \times 4 \times 4} = 12 \ \mathrm{cm}^2$$

Hence, the maximum possible area of this isosceles triangle is 12 cm^2.

Video Solution:

Q10 Text Solution:



Let's draw AP perpendicular to BC, so now we have two right-angle triangles, ABP and ACP.

Now, since, $\angle ABC = 45^\circ$, so $\angle BAP = 180^\circ - 90^\circ - 45^\circ = 45^\circ$. So, ABP is an isosceles right-angled triangle. i.e., BP = AP = x (say).

Therefore, by the Pythagoras theorem, we get

$$x^2 + x^2 = (4\sqrt{2})^2$$
$$2x^2 = 32$$
$$x^2 = 16$$
$$x = 4 \text{ cm}$$

Thus, AP = BP = 4 cm.

Now, in the right-angled triangle APC

$$AP^2 + PC^2 = AC^2$$
$$\Rightarrow (4)^2 + PC^2 = 5^2$$
$$\Rightarrow PC = \sqrt{25 - 16} = 3\ cm$$

So area of the triangle ABC =

$$\frac{1}{2} \times AP \times BC = \frac{1}{2} \times 4 \times (BP + PC)$$
$$= 2 \times (4 + 3) = 14 \text{ cm}^2$$

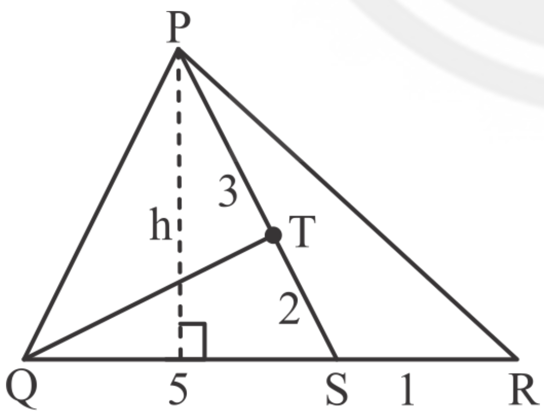
Video Solution:

Q11 Text Solution:

According to the given condition,

QS : SR = 5 : 1

Let h be the height of the triangle PQR. Then, it will also be the height of the triangles PQS and PSR.



Thus, the ratio of areas of the triangles,

$$Area\ PQS\ :\ Area\ PSR = \frac{1}{2} \times 5 \times h$$
$$: \frac{1}{2} \times 1 \times h = 5 : 1$$

Similarly, since, PT : TS = 3 : 2, then the ratio of areas of the triangles,

$Area\ PTQ\ :\ Area\ TQS\ = 3 : 2$.

Now, the area of the triangle PQR is given as 36 cm^2. Also, the area of the PQR is the sum of the areas of the triangles PQS and PSR.

Therefore, (5 + 1) = 6 units = 36 cm^2

1 unit = 6 cm^2

5 units = 5(6) = 30 cm^2 . Thus, the area of

$\triangle\ PQS$ = 30 cm^2.

So, the area of the

$$\triangle\ PQT = \frac{3}{3+2} \times 30 = 18\ cm^2$$

Video Solution:

Q12 Text Solution:

Let's set up the conditions. We're given that

a – b = 4 ...(1)

a + b + c = 24 ...(2)

From (1) and (2), we get

(b + 4) + b + c = 24

2b + 4 + c = 24 c = 20 – 2b ...(3)

For a triangle, the side lengths must satisfy the triangle inequalities. One of these (b + a > c) becomes

b + (b + 4) > 20 – 2b 4b > 16 b > 4 ...(4),

and another (b + c > a) simplifies to

b + (20 – 2b) > b + 4 20 – b > b + 4 20 > 2b + 4 b < 8 ...(5)

i.e., 4 < b < 8. ...(6)

So, using (1), (3) and (6), the possible integer pairs of (a, b, c) are: (9, 5, 10), (10, 6, 8), and (11, 7, 6).

Now, since, the triangle is obtuse, so it must satisfy the obtuse triangle inequality, i.e., the square of the largest side must be greater than the sum of the squares of the smaller ones.

For (9, 5, 10), $10^2 \not> 9^2 + 5^2$, so this pair cannot be possible.

For (10, 6, 8), $10^2 \not> 6^2 + 8^2$, so this pair is also cannot be possible.

For (11, 7, 6), $11^2 > 7^2 + 6^2$, this is the only possible pair.

Now, to find the area we use Heron's formula.

The semi-perimeter is

s = $\frac{11+7+6}{2}$ = 12.

Then the area of the triangle is:

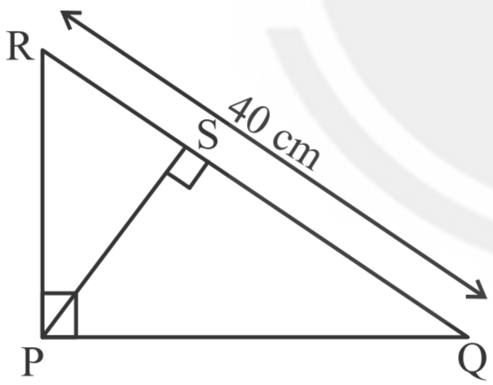
= $\sqrt{12(12-11)(12-7)(12-6)}$

= $\sqrt{12 \times 1 \times 5 \times 6}$

= $6\sqrt{10}$.

Video Solution:

Q13 Text Solution:



PS can be maximum when PQR is an isosceles right-angled triangle. i.e., when PR = PQ.

So, from the right-angled triangle PQR, we get

$PQ^2 + PR^2 = QR^2$

$2PQ^2 = QR^2$

$PQ = \sqrt{\frac{QR^2}{2}} = \sqrt{\frac{40^2}{2}} = \sqrt{800} = 20\sqrt{2}$ cm

So, the area of triangle PQR =

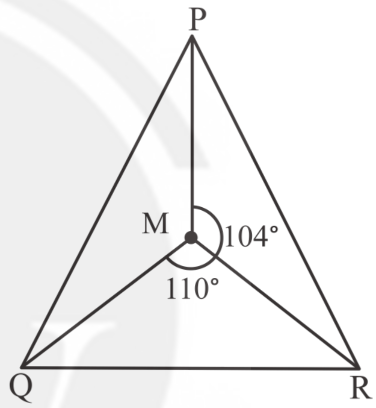
$\frac{1}{2} \times PR \times PQ = \frac{1}{2} \times QR \times PS \Rightarrow$

$PS = \frac{20\sqrt{2}\times 20\sqrt{2}}{40} = 20 \text{ cm}$

i.e., the maximum possible length of PS = 20 cm.

Video Solution:

Q14 Text Solution:



$\angle \mathrm{PMQ} + \angle \mathrm{PMR} + \angle \mathrm{QMR} = 360^\circ$

$\Rightarrow \angle \mathrm{PMQ} = 360^\circ - 104^\circ - 110^\circ = 146^\circ$

Also

$\angle \mathrm{PMQ} = 90^\circ + \frac{\angle \mathrm{PRQ}}{2}$ $\left(\text{Since M is the incenter}\right)$

So

$146^\circ = 90^\circ + \frac{\angle \mathrm{PRQ}}{2}$

$\Rightarrow \angle \mathrm{PRQ} = 2\left(56^\circ\right)$

$\Rightarrow \angle \mathrm{PRQ} = 112^\circ$

Video Solution:

Q15 Text Solution:

We know that the ratio of lengths of Circumradius and Inradius of an equilateral triangle is 2 : 1

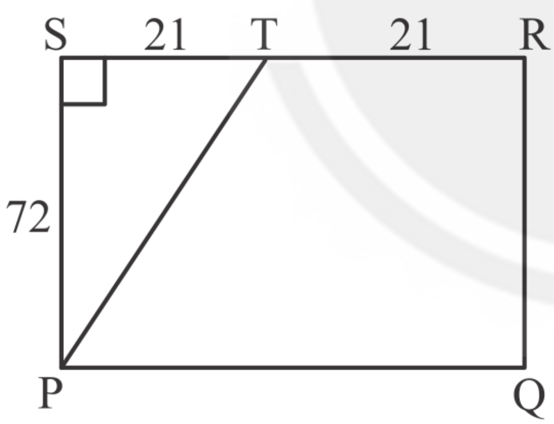
Thus, the ratio of areas of circumcircle and incircle will be 4 : 1

Given that Area of circumcircle = 100 sq. units

Area of incircle = 100 x $\frac{1}{4}$ = 25 sq. units

Video Solution:

Q16 Text Solution:



As PQ = 42, thus RS = 42

As T is the midpoint of RS, TS = $\frac{42}{2}$ = 21

Also, QR = PS = 72

Further, $\angle$RSP = 90

Therefore, PST becomes a right triangle

TS = 3 x **7,**

PS = 3 x **24,**

PT = 3 x **25** = 75 (7, 24, 25 is a pythagorean triplet)

Inradius (radius of incircle) of a right triangle = $\frac{\text{Perpendicular } + \text{Base } - \text{ Hypotenuse}}{2}$

= $\frac{21 + 72 - 75}{2}$

= **9**

Video Solution:

Q17 Text Solution:

Taking $XA = 6a$
$AN = 2a$
$XN = 4a$
$XM = AM = 3a$
So
$MN = XN - AM = a$
Since $\triangle AYM$ and $\triangle XYN$ have equal bases and equal height, their areas will be equal.
Let

$$ar\left(\triangle AYM\right) = ar\left(\triangle XYN\right) = x$$

So

$$ar\left(\triangle XYA\right) = 2x$$

and since XA is median to YZ, it divides the area of $\triangle AYM$ and $\triangle XYZ$ equally into two parts, so

$$ar\left(\triangle XYZ\right) = 2 \times ar\left(\triangle XYA\right) = 4x$$

Also since $\triangle YMN$ and $\triangle AYM$ have same altitude, the ratio of their areas will be equal to the ratio of their bases, so

$$\frac{ar(\triangle YMN)}{ar(\triangle AYM)} = \frac{MN}{AM} = \frac{1}{3}$$

So

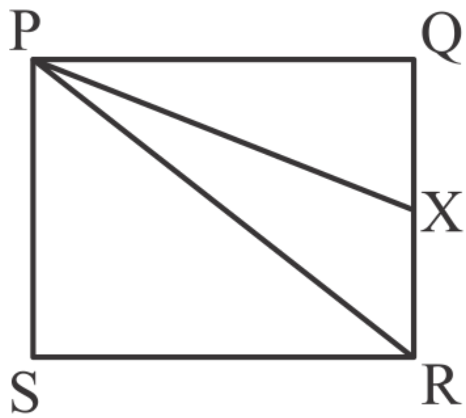
$$ar\left(\triangle YMN\right) = \frac{ar(\triangle AYM)}{3} = \frac{x}{3}$$

So

$$\frac{ar(\triangle XYZ)}{ar(\triangle YMN)} = \frac{4x}{\frac{x}{3}} = 12 : 1$$

Video Solution:

Q18 **Text Solution:**



Opposite sides are divided in the ratio of adjacent sides by an angle bisector.
So

$$\frac{QX}{XR} = \frac{PQ}{PR}$$

Since PR is diagonal of the square, PR =

$$\sqrt{2}PQ = 12\sqrt{2}\ cm$$

So

$$\frac{QX}{XR} = \frac{12}{12\sqrt{2}}$$

$$\Rightarrow QX : XR = 1 : \sqrt{2}$$

So, Area(PRX)=

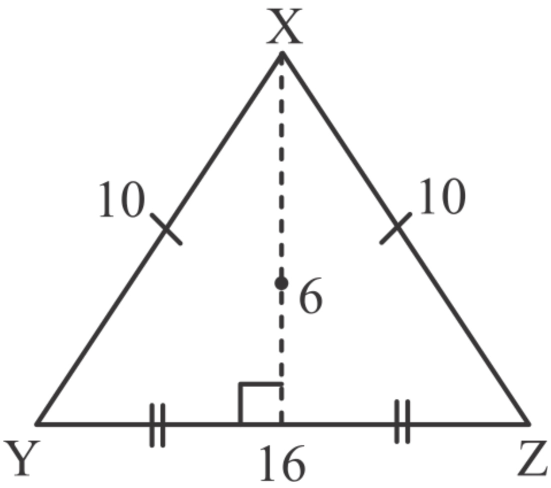
$$\frac{1}{2} \times PQ\left(height\right) \times XR = \frac{1}{2} \times 12$$

$$\times \left(\frac{\sqrt{2}}{1+\sqrt{2}} \times 12\right) = \frac{72\sqrt{2}}{1+\sqrt{2}}$$

$$= 72\sqrt{2}\left(\sqrt{2}-1\right) = 72\left(2-\sqrt{2}\right)\ cm^2$$

Video Solution:

Q19 **Text Solution:**



Point equidistant from X, Y and Z will be the circumcenter of the triangle XYZ, so we need to find the circumradius here.

Since XYZ is an isoceles triangle, if we drop a perpendicular from X on YZ, it will bisect YZ into two equal parts of 8 km each.

So looking at the half triangles we can see the pythagorean triplet $\left(6,\ 8,\ 10\right)$.

So the altitude through X $=$ 6 km

So $ar\left(\triangle XYZ\right) = \frac{1}{2} \times 6 \times 16 = 48$ km^2

So

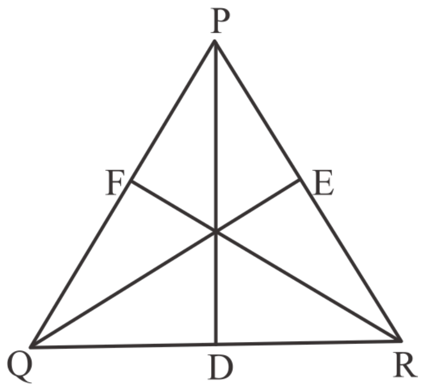
$\text{Area} = \frac{a \times b \times c}{4R}$

$\Rightarrow R = \frac{a \times b \times c}{4(\text{Area})} = \frac{10 \times 10 \times 16}{4 \times 48} = \frac{25}{3}$

$= 8\frac{1}{3}\text{km}$

Video Solution:

Q20 Text Solution:



By Apollonius theorem

$PQ^2 + PR^2 = 2\left(PD^2 + QD^2\right)$

$QP^2 + QR^2 = 2\left(QE^2 + RE^2\right)$

$RP^2 + RQ^2 = 2\left(RF^2 + PF^2\right)$

So

$2\left(PQ^2 + RP^2 + QR^2\right) = 2\left(PD^2 + QD^2 + QE^2 + RE^2 + RF^2 + PF^2\right)$

$\Rightarrow PQ^2 + RP^2 + QR^2 = PD^2 + QD^2 + QE^2 + RE^2 + RF^2 + PF^2$

$\Rightarrow 10^2 + 12^2 + 16^2 = PD^2 + QE^2 + RF^2 + 5^2 + 6^2 + 8^2$

$\Rightarrow PD^2 + QE^2 + RF^2 = 100 + 144 + 256 - 25 - 36 - 64$

$\Rightarrow PD^2 + QE^2 + RF^2 = 500 - 125$

$\Rightarrow PD^2 + QE^2 + RF^2 = 375$

Alternatively,

$PD^2 + QE^2 + RF^2 = \frac{3}{4} \times \left[PQ^2 + QR^2 + PR^2\right]$

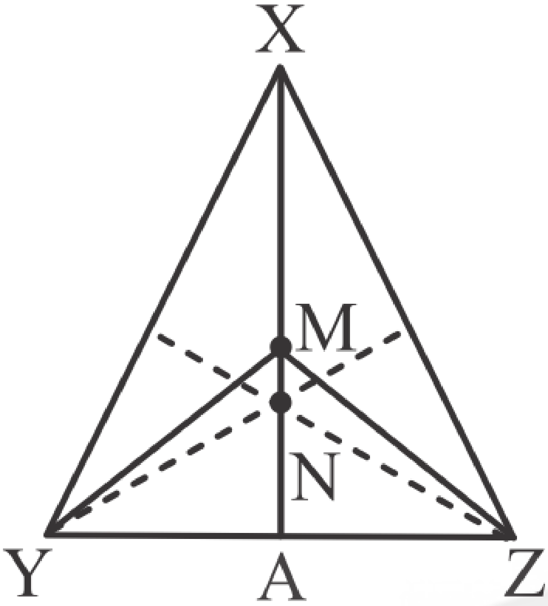
$PD^2 + QE^2 + RF^2 = \frac{3}{4} \times \left[100 + 144 + 256\right]$

$PD^2 + QE^2 + RF^2 = \frac{3}{4} \times 500$

$PD^2 + QE^2 + RF^2 = \mathbf{375}$

Video Solution:

Q21 Text Solution:



Let XA be the median through X, so

$YA = AZ$

So

$ar(\triangle XYA) = ar(\triangle XZA) = \frac{1}{2} ar(\triangle XYZ)$

Also

$XM = MA$

So

$ar(\triangle YMA) = \frac{1}{2} ar(\triangle XYA) = \frac{1}{4} ar(\triangle XYZ)$

and

$ar(\triangle ZMA) = \frac{1}{2} ar(\triangle XZA) = \frac{1}{4} ar(\triangle XYZ)$

So

$ar(\triangle MYZ) = ar(\triangle ZMA) + ar(\triangle YMA) = \frac{1}{2} ar(\triangle XYZ)$

So

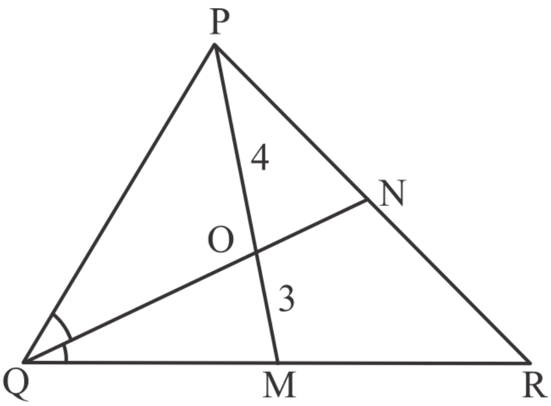
$ar(\triangle XYZ) = 2\, ar(\triangle MYZ) = 30$ square units

Since three medians divide a triangle into 6 triangles of equal areas, so

$ar(\triangle NYZ) = ar(\triangle YNA) + ar(\triangle ZNA) = \frac{1}{6}(\triangle XYZ) + \frac{1}{6}(\triangle XYZ) = \frac{1}{3} \times 30 = 10$ square units

Video Solution:

Q22 Text Solution:



In $\triangle PQM$, QO is the angle bisector of angle PQM, so

$\frac{PO}{OM} = \frac{PQ}{QM}$

$\Rightarrow \frac{PQ}{QM} = \frac{4}{3}$

Let

$PQ = 4x$

and $QM = 3x$

Since PM is median,

$QM = MR$

So

$QR = 6x$

So using angle bisector theorem in $\triangle PQR$

$\frac{PN}{NR} = \frac{QP}{QR} = \frac{4x}{6x} = \frac{2}{3}$

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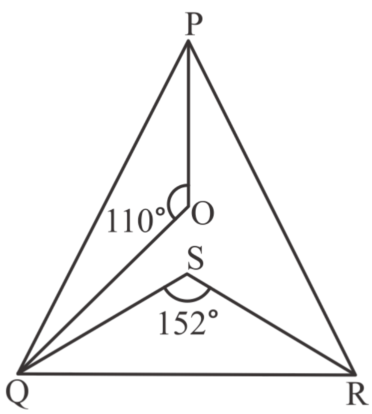
Q23 Text Solution:



$\angle POQ = 110^\circ$ is at orthocentre, so

$\angle POQ + \angle PRQ = 180^\circ$

$\Rightarrow \angle PRQ = 180^\circ - 110^\circ = 70^\circ$

Also $\angle QSR = 152^\circ$ is at circumcentre, so

$\angle QSR = 2 \times \angle QPR$

$\Rightarrow \angle QPR = \frac{152^\circ}{2} = 76^\circ$

So

$\angle QPR + \angle PRQ + \angle RQP = 180^\circ$

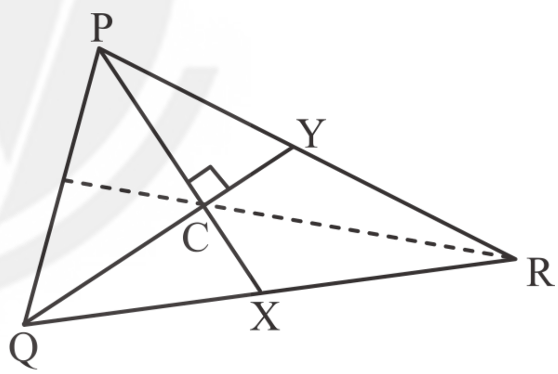
$\Rightarrow \angle RQP = 180^\circ - 76^\circ - 70^\circ = 34^\circ$

Since $\angle PIR$ will lie at incentre

$\angle PIR = 90^\circ + \frac{\angle RQP}{2} = 90^\circ + 17^\circ = 107^\circ$

Video Solution:

Q24 Text Solution:



Let C be the centroid of the triangle where the three medians intersect. The three medians of a triangle divide it into 6 triangles of equal areas, so

$\text{ar}(\triangle PCY) = \frac{1}{6}\text{ar}(\triangle PQR) = \frac{156}{6}$

$= 26$ sq units

Also

$\text{ar}(\triangle PCY) = \frac{1}{2} \times PC \times CY$

$PC : CX = 2 : 1$

$\Rightarrow PC = \frac{2}{3}PX$

Also

$QC : CY = 2 : 1$

$\Rightarrow CY = \frac{1}{3}QY = 6$ units

So

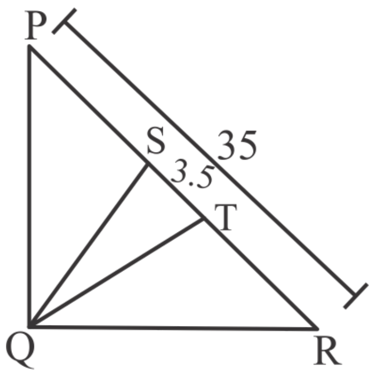
$\frac{1}{2} \times PC \times CY = 26$

$\Rightarrow \frac{1}{2} \times \frac{2}{3}PX \times 6 = 26$

$\Rightarrow PX = 13$ units

Video Solution:

Q25 Text Solution:



$PT = RT = \frac{PR}{2} = 17.5$ cm

So

$PS = PT - ST = 14$ cm

and

$RS = RT + ST = 21$ cm

So

$\frac{PQ}{QR} = \frac{PS}{RS}$

$\Rightarrow \frac{PQ}{QR} = \frac{14}{21}$

$\Rightarrow \frac{PQ}{QR} = \frac{2}{3}$

Let $PQ = 2x$ cm and $QR = 3x$ cm.

So

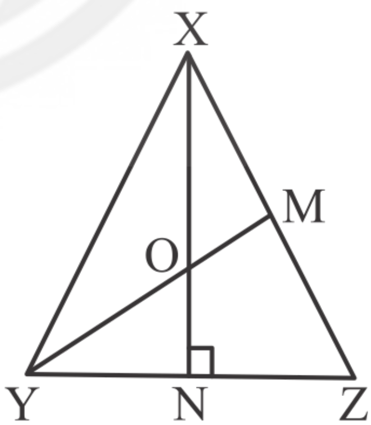
$(2x)^2 + (3x)^2 = (35)^2 \Rightarrow x^2 = \frac{1225}{13}\ \text{cm}^2$

So area of triangle $PQR = \frac{1}{2} \times PQ \times QR = \frac{1}{2} \times 2x \times 3x = 3x^2 = \frac{3 \times 1225}{13}$

$= \frac{3675}{13} = 282.69 = 283\ \text{cm}^2$

Video Solution:

Q26 Text Solution:



Let YM be the median and XN be the perpendicular and let them intersect at O.

So

$YM = 19.5$ units and $XN = 15$ units

Perpendicular to the unequal side of an isoceles triangle is also a median to that side.

So O is the centroid of the triangle . (Intersection point of medians)

So

$YN = ZN$

Also

$OY = \frac{2}{3}YM = 13$ units

$ON = \frac{1}{3}XN = 5$ units

So in right angle triangle ONY

$OY^2 = ON^2 + YN^2$

$\Rightarrow 169 = 25 + YN^2$

$\Rightarrow YN = 12$ units

And since $YN = NZ$,

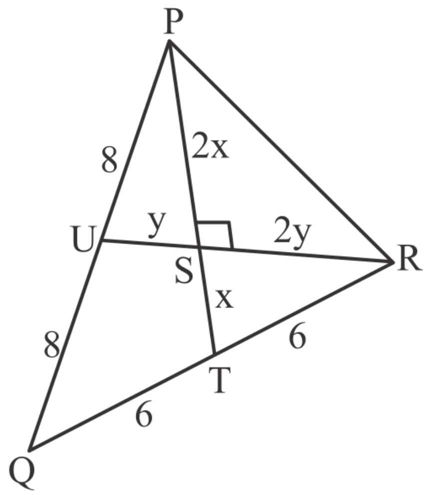
$YZ = 24$ units

So

$ar(\triangle XYZ) = \frac{1}{2} \times YZ \times XN = \frac{1}{2} \times 24 \times 15 = 180$ square units

Video Solution:

Q27 Text Solution:



Let the medians be PT and RU and let them intersect at S.

Since S is centroid, $RS : SU = 2 : 1$ and $PS : ST = 2 : 1$

Let

$PS = 2x$, $ST = x$, $RS = 2y$ and $SU = y$

. Since the medians intersect at right angles, triangles RST and PSU are right angle traingles.

So

$RT^2 = RS^2 + ST^2$ and $PU^2 = PS^2 + SU^2$

Since PT and RU are medians they will bisect QR and PQ respectively.

So

$RT = QT = 6$ units

$PU = QU = 8$ units

So

$6^2 = (2y)^2 + (x)^2 \Rightarrow x^2 + 4y^2 = 36$

and

$8^2 = (2x)^2 + (y)^2 \Rightarrow 4x^2 + y^2 = 64$

Adding the two equations

$5\left(x^2 + y^2\right) = 100$

$\Rightarrow x^2 + y^2 = 20$

Since

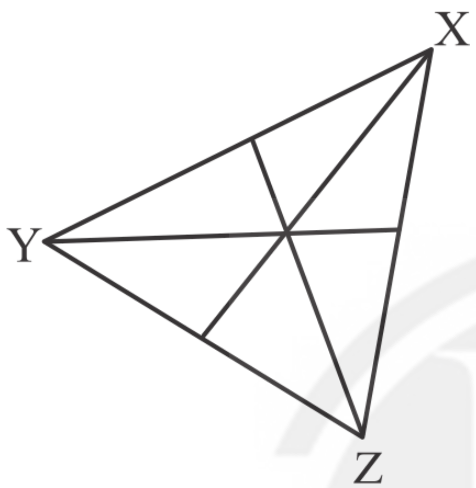
$PR^2 = (2x)^2 + (2y)^2 \Rightarrow PR^2 = 4\left(x^2 + y^2\right)$

$\Rightarrow PR = \sqrt{80}$

$\Rightarrow PR = 4\sqrt{5}$ units

Video Solution:

Q28 Text Solution:



The three medians of triangle divide it into 6 non overlapping triangles of equal area, so

MXZ will be formed by two such triangles, so ar(MXZ)=

$\frac{1}{3} \times ar\left(\triangle XYZ\right) = \frac{1}{3} \times \frac{\sqrt{3}}{4} \times \left(4\sqrt{3}\right)^2$

$= 4\sqrt{3}\ cm^2$

MX=MY=MZ = $\frac{Side}{\sqrt{3}} = \frac{4\sqrt{3}}{\sqrt{3}} = 4\ cm$

So semiperimeter of MXZ =

$\frac{4+4+4\sqrt{3}}{2} = \left(4 + 2\sqrt{3}\right)\ cm$

So

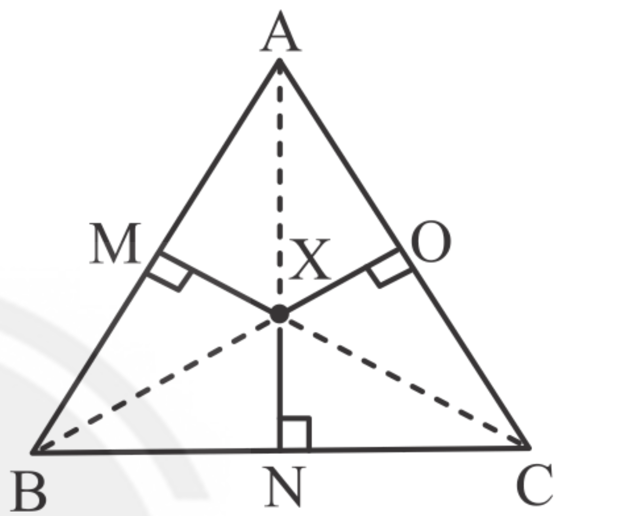
Inradius of MXZ =

$\frac{Area}{semiperimeter} = \frac{4\sqrt{3}}{4+2\sqrt{3}} = \frac{2\sqrt{3}}{2+\sqrt{3}} =$

$\left(4\sqrt{3} - 6\right)\ cm$

Video Solution:

Q29 Text Solution:



Let ABC be an equilateral triangle with X lying inside such that MX, NX and OX are the perpendiculars to AB, BC and CA.

If we consider the side length of the triangle be a, then

$\frac{\sqrt{3}}{4}a^2 = \frac{1}{2}\left(a\right)\left(MX\right) + \frac{1}{2}\left(a\right)\left(NX\right)$

$+ \frac{1}{2}\left(a\right)\left(OX\right)$

$\Rightarrow \frac{\sqrt{3}}{4}a^2 = \frac{1}{2}a\left(MX + NX + OX\right)$

$\Rightarrow a = \frac{2\times 8\sqrt[4]{3}}{\sqrt{3}}$

So

$ar\left(\triangle ABC\right) = \frac{\sqrt{3}}{4}a^2$

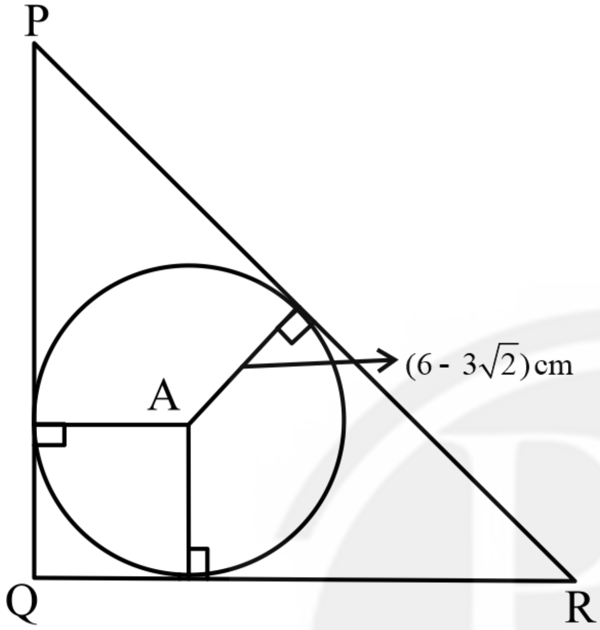
$= \frac{\sqrt{3}}{4}\left(\frac{2\times 8\sqrt[4]{3}}{\sqrt{3}}\right)^2$

$= \frac{\sqrt{3}}{4} \times \frac{4\times 64\sqrt{3}}{3}$

$= 64\ cm^2$

Video Solution:

Q30 Text Solution:



If PQR is a right angled isosceles triangle then it is a 45:45:90 triangle, which means its sides are in ratio

PQ:QR:PR=$1 : 1 : \sqrt{2}$

The point inside the circle equidistant from all the three sides is the incenter of the circle, so the inradius = $(6 - 3\sqrt{2})\ cm$

Let the sides be x, x, $\sqrt{2}$x, so

r=$\frac{a+b-c}{2}$

$\Rightarrow 6 - 3\sqrt{2} = \frac{x+x-\sqrt{2}x}{2}$

$\Rightarrow x = \frac{2(6-3\sqrt{2})}{2-\sqrt{2}}$

$\Rightarrow x = 6\ cm$

So area of triangle PQR = $\frac{1}{2} \times 6 \times 6$ = 18 cm^2

Video Solution: